with sufficiently high temporal resolution and with complete spatially averaged rather than point values.
Remote sensing from satellites can deliver information on GHG soil emissions by estimating tropospherical, near-surface CO_2 and CH_4 concentrations IR Infrared portion of the spectrum. Therefore all the statements are correct.

74. In the context of which one of the following are the terms 'pyrolysis and plasma gasification' mentioned?

(a) Extraction of rare earth elements.
(b) Natural gas extraction technologies.
(c) Hydrogen fuel-based automobiles.
(d) Waste-to-energy technologies.

Ans : (d) Pyrolysis is a process by which chemical substances are decomposed in the absence of oxygen.
Plasma gasification is an extreme thermal process by which plasma is converted into organic gas (hydrogen, carbon monoxide), by using plasma.
Electricity/heat is generated through combustion by waste-to-energy technology. It can refer to methane, methanol, ethanol or synthetic fuel plasma gasification in the context of waste-to-energy technology. Clear energy is produced by recycling waste through pyrolysis and plasma gasification.

75. Consider the following statements :
A digital signature is

1. **an electronic record that identifies the certifying authority issuing it.**
2. **used to-serve as a proof of identity of an individual to access information server on Internet.**
3. **an electronic method of signing an electronic document and ensuring that the original content is unchanged.**

Which of the statements given above is/are correct?

(a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans : (c) Digital signature is an electronic method of signing an electronic document and ensuring that the original content is unchanged. The digital signatures are often used to implement electronic signatures. A valid digital signature assumes the recipient that the message has been generated by a known sender one has not been altered in transit. The above statement (3) is correct.

76. With reference to communication technologies, what is/are the difference/differences between LTE (Long-Term Evolution) and VoLTE (Voice over Long-Term Evolution)?

1. **LTE is commonly marketed as 3G and VoLTE is commonly marketed as advanced 3G.**
2. **LTE is data-only technology and VoLTE is Voice-only Technology.**

Select the correct answer using the code given below.

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans : (d) LTE stands for long-term Evolution. It provides 4G network service and not the 3G network service. The launch of LTE service in India was started by the service provider company Airtel. It works on voice-only technology and not on data only technology.
VoLTE stands for voice over long-term Evolution. It also works solely on 4G technology. The VoLTE service was launched by Reliance company. VoLTE works on voice-over technology and not voice- only technology, so statement (2) is incorrect.
Note- LTE is commonly marketed as 4G while VoLTE is commonly marketed as advanced 4G.

ENVIRONMENT & ECOLOGY

77. In India, 'extended producer responsibility' was introduced as an important feature in which of the following?

(a) The Bio-medical Waste (Management and Handing) Rules, 1998.
(b) The Recycled Plastic (Manufacturing and Usage) Rules, 1999.
(c) The e-Waste(Management and Handing) Rules, 2011.
(d) The Food Safety and Standard Regulations, 2011.

Ans : (c) The e-waste (Management and Handling) Rules, 2011 recognizes producers liability for recycling and reducing e-waste in the country.
Extended producer responsibility (EPR) is a policy approach under which producers are given a significant responsibility-financial and/ or physical for the treatment or disposal of post-consumer products.
The central pollution control Board (CPCB) has been given the Extended producer responsibility (EPR) authorization under the new e-waste rules.

78. In the context of proposals to the use of hydrogen-enriched CNG (H-CNG) as fuel for buses in public transport, consider the following statements :

1. **The main advantage of the use of H-CNG is the elimination of carbon monoxide emissions.**
2. **H-CNG as fuel reduces carbon dioxide and hydrocarbon emissions.**
3. **Hydrogen up to one fifth by volume can be blended with CNG as fuel for buses.**
4. **H-CNG makes the fuel less expensive than CNG.**

Which of the statements given above is/are correct?

(a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans : (b) The use of H-CNG as a fuel for buses by combining hydrogen with CNG is at an experimental level. Its use can reduce carbon dioxide & Hydro carbon emission but not eliminate carbon monoxide emissions by upto 70 percent. Simultaneously, this fuel reduces carbon dioxide and Hydrocarbon emissions Hydrogen can be mixed with CNG upto 18 percent, which is equivalent to one-fifth of this fuel. However, H-CNG is more expensive than CNG because both its price and storage are expensive. Hence both (2) and (3) are correct in the above statements.

79. Consider the following statements :

1. **Under Ramsar Convention, it is mandatory on the part of the Government of India to protect and conserve all the wetlands in the territory of India.**
2. **The Wetlands (Conservation and Management) Rules, 2010 were framed by the government of India based on the recommendations of Ramsar Convention.**
3. **The Wetlands (Conservation and Management) Rules. 2010 also encompass the drainage area or catchment regions of the wetlands as determined by the authority.**

Which of the statements given above is/are correct

(a) 1 and 2 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans : (c) The Ramsar convention is an intergovernmental treaty for the conservation of wetlands. When a country joins the Ramsar convention, it gets listed in the list of the countries that signed the treaty. Once a country joins the treaty, it has some commitments that it has to fulfill as obligations statement, (1) of the above statements is incorrect, because according to the Ramsar Convention, it is not binding on the Government of India to protect and preserve all the wetlands in the territory of India.

80. Consider the following statements :

1. **As per recent amendment to the Indian Forest Act, 1927, forest dwellers have the right to fell the bamboos grown on forest areas.**
2. **As per the Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, bamboo is a minor forest produce.**
3. **The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 allows ownership of minor forest produce to forest dwellers.**

Which of the statements given above is/are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans : (b) The Amendment Act removes bamboo as a tree from section 2 (7) of India forest Act, 1927 which would ease transit and feeling restrictions on bamboo in private lands; moreover the Act still has legal provisions unauthorized extraction of bamboo from the forest areas and ampoules forest departments to prosecute any person or agency found to be doing so therefore, statement (1) is incorrect.
Minor forest produce is defined under the scheduled tribes and other traditional forest Dwellers (Recognition of forest rights) Act, 2006 as all non-timber forest produce of plant origin and includes bamboo, brushwood, stumps, cares , Tusser, cocoon, honey waxes, Lac, tendu/Kendu, leaves, medicinal plants and herbs, roots tuber, etc. Therefore, statement (2) is correct.
The Forest Rights Acts, 2006 defines forest, rights as inclusive of right of ownership, access to collect, use and dispose of minor forest produce which have traditionally been collected within or outside village boundaries: Individuals, communities and gram shabhas having rights under this particular section of the Act will not only have the rights to use but also rights of ownership over MFPs. Therefore, statement (3) is correct.

81. Which one of the following National Parks lies completely in the temperate alpine zone?

(a) Manas National Park
(b) Namdapha National Park
(c) Neora Valley National Park
(d) Valley of Flowers National Park

Ans : (d) The alpine forests are found all along the Himalayas at an altitude ranging between 2500 m to 3500 m.
Valley of flowers National Park- At 3352 to 3658 meters above level.
Namdapha National Park has a very wide altitude variation from 200 to 4500 m in the snow-capped mountain. Thus, it does not lie completely in the temperate alpine zone hence the correct option is (d).

82. Consider the following statements :

1. Asiatic lion is naturally found in India only.
2. Double-humped camel is naturally found in India only.
3. One-horned rhinoceros is naturally found in India only.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (a) Double- humped camel, which is also called Bactrian camel, is found in the Ladakh region of India & also in the Gobi desert, Mongolia, Kazakhstan and some parts of China. One-Horned rhinoceros is commonly found in Nepal, Bhutan, Pakistan and Assam in India, while the Asiatic lion is naturally found only in the Gir National Park of India. The Asiatic lion is included in the Red Data book of the IUCN, it is noted that, the Red Data List, include threatened animals. The Asiatic lion is protected under the wildlife protection Act, 1972 so the correct answer is option (a).

83. Which of the following are in Agasthyamala Biosphere Reserve?

(a) Neyyar, Peppara and Shendurney Wildlife Sanctuaries: and Kalakad Mundanthura Tiger Reserve
(b) Mudumalai , Sathyamangalam and Wayanad Wildife Sanctuaries; and Silent Valley National Park
(c) Kaundinya, Gundla Brahmeswaram and Papikonda Wildlife Sanctuaries: and Mukurthi National Park
(d) Kawal and Sri Venkateswara Wildlife Sancturaries: and Nagarjunasagar-Srisailam Tiger Reserve